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Image Acquisition and Analysis in Neuroscience

Assignment Sheet 5

Solution has to be uploaded by Januar 8, 2026, 10:00 a.m., via eCampus

If you have questions concerning the exercises, please use the forum on eCampus.

- Please work on this exercise in **small groups** of 3 students. Submit each solution only once, but clearly indicate who contributed to it by forming a team in eCampus. Remember that all team members have to be able to explain all answers.
- Please submit your answers in PDF format, and your scripts as *.py/*.ipynb files. If you are using Jupyter notebook, please also export your scripts and results as PDF.
- Admission to the exam requires **at least 75 points** (50% of the regular points from all six sheets) and presenting an answer in class **at least once.**

Exercise 1 (Tversky Loss for Image Segmentation, *5 Points*)

The Tversky Index unifies and generalizes the Dice and IoU scores. It is given by

$$\frac{|A \cap B|}{|A \cap B| + \alpha |A \setminus B| + \beta |B \setminus A|}$$

with $\alpha \geq 0$, $\beta \geq 0$.

a) Find the values of α and β that yield the Dice and the IoU scores, respectively. (2P)

b) In Chapter 5.7, we discussed the Dice loss for training segmentation networks. Write down a corresponding Tversky loss that is based on the Tversky Index, but is a differentiable function of probabilistic segmentation outputs a_i and ground truth labels b_i. (1P)

c) Describe a case in which it is beneficial to use a Tversky loss with values α and β different from the ones in a). (1P)

d) The focal Tversky loss is based on the regular Tversky loss, but raises it to a power of $\gamma \geq 1$. Explain the effect of using the focal Tversky loss with $\gamma > 1$. (1P)

Exercise 2 (Statistical Testing, *10 Points*)

Assume that a new drug has been proposed for the treatment of bipolar disorder. You would like to investigate whether treatment with this drug has an effect on amygdala volume. These volumes are determined based on brain MR segmentations, and are given in `bipolar.xls`. For each subject in the randomized study, this spreadsheet contains a measurement at the time of diagnosis, a follow-up measurement (six months later), and an indicator of whether treatment involved the new drug.

a) What is a suitable null hypothesis? Should we use a one-sided or a two-sided test? Briefly justify why. (2P)

b) Use a suitable variant of the t-test to test your null hypothesis. We recommend using `scipy` for this. Note that `scipy` offers predefined functions for t-tests, you do not have to implement the individual steps. Does your test reject the null hypothesis at $\alpha = 0.05$? If so, did the drug increase or decrease amygdala volumes on average? (3P)

c) Use a suitable variant of ANOVA (F-test) to test the same null hypothesis. Again, you can use a predefined `scipy` function. How is the resulting p value related to the one from the t-test? Briefly state why. (2P)

d) Repeat the t-test from b), but compute the p value with a permutation-based test. Compute p values 100 times each based on 100, 1000, and 3000 random permutations. Create three side-by-side boxplots of the respective p values and describe your observations. In this example, is it computationally feasible to perform all possible permutations? Justify your answer. (3P)

   Hint: Permutation-based testing is an option in `scipy`'s implementation of t-tests as of version 1.7.0. You do not have to implement it yourself.

Exercise 3 (Threshold-Free Cluster Enhancement, *10 Points*)

In this exercise, you will implement a 2D version of the Threshold-Free Cluster Enhancement (TFCE) algorithm and apply it to the statistical parametric map in `spm.npy` which you can download from eCampus. The 2D map can be seen as an image with a t score at each pixel. Since t scores are signed floating-point numbers, we store it in numpy's `.npy` format rather than in a traditional image format.

a) Write a function that, for any given threshold value, creates a binary mask and a label image. Labeling means that you should assign a unique identifier to each connected component of the mask. In the label image, the value of each pixel should be the identifier of the component it belongs to. Feel free to make use of the functionality in `scipy.ndimage`. (3P)

b) Write a function that, given the label image, creates an extent image which, for each pixel, should contain the extent (i.e., size in number of pixels) of the component it belongs to. (3P)

c) Write a loop that iterates over a given number of threshold values (between 0 and the maximum found in the input) and, for each pixel p, performs numerical integration of the TFCE integral

$$TFCE(p) = \int_{h=0}^{h_p} e(h)^E h^H \, dh$$

   with the commonly used values $E = 0.5$, $H = 2$. (3P)

d) Try different step sizes for the discretization of the integral (i.e., different number of threshold values). How does the resulting image change? (1P)

Please submit your script and the TFCE-transformed input for at least two different discretizations.

Good Luck!